



Finding the n^{th} term

Task A: Finding the n^{th} term of increasing sequences

1) Find the n^{th} term of the sequences below:

a) $10, 17, 24, 31, 38, \dots$ $7n + 3$

b) $1, 8, 15, 22, 29 \dots$ $7n - 6$

c) $-10, -7, -4, -1, 2 \dots$ $3n - 13$

d) $0.9, 1.3, 1.7, 2.1, 2.5 \dots$ $0.4n + 0.5$

e) $\frac{1}{4}, \frac{3}{4}, \frac{5}{4}, \frac{7}{4}, \frac{9}{4}, \dots$ $\frac{1}{2}n - \frac{1}{4}$

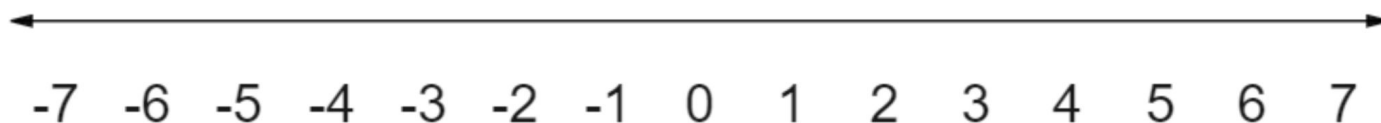
2) Andeep says:

The sequence $-5, -3, -1, 1, 3, \dots$ is $2n - 5$ because the common difference is $+2$ and it starts at -5



Give two explanations as to why Andeep is wrong.

The number line may help.



The sequence $2n$ starts: $2, 4, 6, \dots$
 $\begin{array}{ccc} -7 & -7 & -7 \\ -5, -3, -1, \dots \end{array}$

Your sequence is a translation by -7 therefore it is $2n - 7$

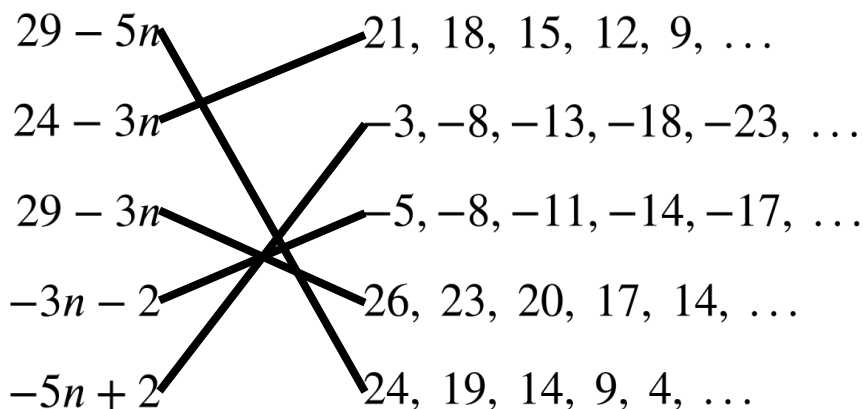
$$n = 1$$

$$2(1) - 5 = -3$$

The first term of the sequence $2n - 5$ is -3 whereas your first term is -5 so $2n - 5$ can not describe your sequence.

Task B: Finding the n^{th} term of decreasing sequences

1) Match the sequences to their n^{th} term.



2) Find the n^{th} term of these sequences.

a) 18, 6, -6, -18, -30 ...

$30 - 12n$

b) -9, -21, -33, -45, -57 ...

$-12n + 3$

c) -15, -18, -21, -24, -27 ...

$-3n - 12$

d) 12, 9, 6, 3, 0 ...

$15 - 3n$

e) 1.2, 0.9, 0.6, 0.3, 0 ...

$1.5 - 0.3n$

f) $3\frac{1}{2}$, 3, $2\frac{1}{2}$, 2, $1\frac{1}{2}$, ... $4 - \frac{1}{2}n$

3) Two students are discussing arithmetic sequences.



Aisha

If you give me any two terms of an arithmetic sequence I can tell you the n^{th} term.

You're right Aisha! If we had 30 and 25 we'd know it's 30, 25, 20, 15, 10, ... with an n^{th} term of $35 - 5n$



Sofia

Write a sentence to explain if Aisha and Sofia are right.

You might have written:

You need to know which two terms you have. If 30 and 25 are terms one and two then yes, it's $35 - 5n$.

But, if 30 and 25 are the fifth and sixth terms, then it's a different sequence altogether.



Task C: Finding and using the n^{th} term

1) Find the n^{th} term, then use approximation and counting on to find the term 705 in the arithmetic sequence starting $-25, -15, -5, 5, 15, \dots$

$$10n - 35 \quad n = 70 \quad 10(70) - 35 = 665$$

$\dots, 665, 675, 685, 695, 705, \dots$
 70th 71st 72nd 73rd 74th

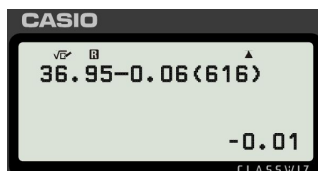
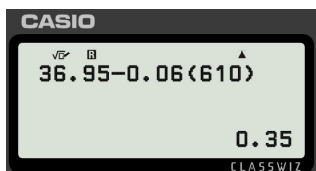
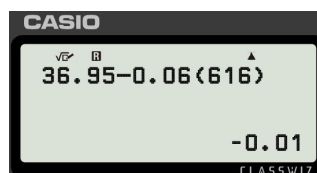
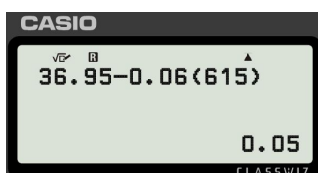
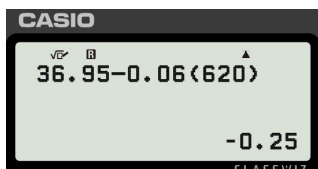
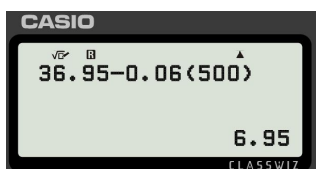
705 is the 74th term in the sequence.

2) Find the n^{th} term, then use approximation and counting on to find the term -554 in the arithmetic sequence starting $-2, -14, -26, -38, -50, \dots$

$$10 - 12n \quad n = 50 \quad 10 - 12(50) = 10 - 600 = -590$$

$\dots, -554, -566, -578, -590, \dots$
 47th 48th 49th 50th *-554 is the 47th term in the sequence.*

3) Jacob wants to know the last term positive term in the sequence $36.95 - 0.06n$. Use trial and improvement on your calculator to help him.



The 615th term is positive. The 616th term is negative.

Therefore, the 615th is the last positive term.

